# HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

## PROFESSIONAL EXPERIENCE — 3.5 YEARS

**Software Engineer — HCL Technologies, Chennai, India** Sep 2021 – Jan 2025

- Designed and built scalable data pipelines using **Python** and **SQL** to ingest and process multi-source transport datasets, handling over 20 million vehicle pings daily.
- Developed automated data cleaning and validation logic in **Python**, reducing data anomaly rates by approximately **18%** for critical telemetry streams.
- Optimised core data processing pipelines for geospatial analysis and real-time data enrichment, improving query performance by an average of **30%**.
- Collaborated with data scientists and product teams to translate complex transportation logic into efficient engine code, ensuring alignment with business requirements.
- Supported pipeline management and job monitoring using **AWS CloudWatch** and internal tooling, guaranteeing reliable daily processing for downstream products.
- Troubleshot complex data processing issues, performing root-cause analysis and streamlining data ingestion pipelines to enhance system reliability.
- Containerised data processing services using **Docker** and deployed them to an **AWS EC2** environment, configuring basic resource limits for predictable performance.

## SKILLS

**Data Engineering** – Data Pipelines, ETL/ELT, Data Modeling, Data Warehousing, Data Quality, Geospatial Analysis, Batch Processing, Real-time Processing

**Cloud Platforms** – AWS (EC2, S3, Lambda, CloudWatch)

**Programming Languages** – Python, SQL, Java

**Databases** – PostgreSQL, MySQL, MongoDB, Snowflake, BigQuery, Redshift

**Data Processing Tools** – Spark, Airflow, dbt, Kafka

**Containerisation** – Docker, Kubernetes

**Methodologies** – Agile, Scrum

## PROJECTS

### Transport Data Ingestion and Processing Engine

- Engineered a data pipeline using **Python** and **SQL** to ingest vehicle telemetry and passenger demand data, processing approximately **1.5 million records** per hour.
- Implemented automated data validation and deduplication routines in **Python**, ensuring data integrity for over **3 billion** processed passenger journeys annually.
- Optimised data transformation jobs for geospatial mapping and corridor analysis, reducing processing times by up to **22%**.

### Real-time Vehicle Telemetry Pipeline

- Developed a **Python**-based pipeline to process high-velocity vehicle pings, matching demand to telemetry data every **22 seconds**.
- Integrated with **AWS S3** for storing raw and processed telemetry data, ensuring reliable data availability for downstream analytics.
- Designed and implemented REST APIs using **Java Spring Boot** for cross-service data exchange, supporting seamless integration with the primary data platform.

## EDUCATION

**M.Sc. Data Analytics — National College of Ireland, Dublin** Feb 2026

**B.E. Computer Science — SNS College of Technology, Coimbatore, India** 2021

## CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)